$$\begin{aligned}
&\frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0, \\
&\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + P' \mathcal{I}_2) = -\rho' g \boldsymbol{k} + \nabla \cdot \boldsymbol{F}_{\boldsymbol{u}}^{\text{visc}}, \\
&\frac{\partial \rho e'}{\partial t} + \nabla \cdot [(\rho e + P)\boldsymbol{u}] = \nabla \cdot \boldsymbol{F}_e^{\text{visc}},
\end{aligned} \tag{11}$$

which is the form used in all the test cases in Section 6; note that we were not able to find a single mesoscale model in the literature which uses this equation set.

2.4. Viscous stresses

It should be mentioned that only equation set 3 has the true viscous stresses; we use *ad hoc* diffusion terms for sets 1 and 2 merely for convenience. The viscous terms of set 2 would be identical to those of set 3 for the momentum if

$$\nabla \cdot (\mu \rho \nabla \boldsymbol{u}) \rightarrow \nabla \cdot \boldsymbol{F}_{\boldsymbol{u}}^{\text{visc}},$$

where the arrow signifies that the term on the left should be replaced by the one on the right. For the potential temperature, after much algebra, one would arrive at

$$\nabla \cdot (\mu \rho \nabla \theta) \rightarrow -\frac{1}{c_p \pi} \left(\nabla \cdot \left(-\frac{\mu c_p}{Pr} \nabla T \right) + \boldsymbol{F}_{\boldsymbol{u}}^{\text{visc}} \cdot \nabla \boldsymbol{u} \right).$$

These viscous terms can then be used to compute the true stresses of set 1. Specifically, for set 1 we would get for momentum

$$\mu \nabla^2 \boldsymbol{u} \rightarrow \frac{1}{\rho} \nabla \cdot \boldsymbol{F}_{\boldsymbol{u}}^{\text{visc}}$$

and for potential temperature

$$\mu \nabla^2 \theta \rightarrow -\frac{1}{c_p \pi \rho} \left(\nabla \cdot \left(-\frac{\mu c_p}{Pr} \nabla T \right) + \boldsymbol{F}_{\boldsymbol{u}}^{\text{visc}} \cdot \nabla \boldsymbol{u} \right).$$

There are two reasons why we do not use these stresses: the first reason is that for the purposes of intercomparison of our models with other previously published models we adhere to the equations used in Straka et al. [51] which use viscous stresses similar to those that we define for sets 1 (Eq. (1)) and 2 (Eq. (4)). The second reason for not using these stresses (at least for set 2) is that the equations would no longer be in conservation form. Due to these issues alone we recommend set 3 to be used for all future nonhydrostatic atmospheric models. For the purposes of this study it is not so important to use the true viscous stresses for sets 1 and 2 since viscosity is only used for one test case. Since no previous work using set 3 was found, we use the true Navier–Stokes viscosity (Eqs. (9) and (10)). Due to this discrepancy in the viscous stresses between the three equation sets, we expect slight differences in the results; the question is how large will these differences be.

3. Definition of the test cases

We now describe the seven test cases used to validate the numerical models. The set of test cases that we use is based on a list first proposed by Skamarock et al. [47]; we have added two bubble tests to the set. None of the cases uses Coriolis and only the density current test requires viscosity.

For all cases, we shall define the initial conditions in terms of Exner pressure, π, and potential temperature, θ. However, for equation sets 2 and 3 we require density, ρ, and for set 3 total energy, e. Thus the conversion from π and θ to ρ and e is as follows:

$$\rho = \frac{P_0}{R\theta} \pi^{\frac{c_v}{R}}$$

and

$$e = c_v \theta \pi + \frac{1}{2}\boldsymbol{u}\cdot\boldsymbol{u} + gz.$$

To simplify the description of the initial conditions we shall write the Exner pressure and potential temperature in terms of a mean value plus a perturbation, that is $\pi(\boldsymbol{x},t) = \bar{\pi}(z) + \pi'(\boldsymbol{x},t)$ and $\theta(\boldsymbol{x},t) = \bar{\theta}(z) + \theta'(\boldsymbol{x},t)$ where the reference state is in hydrostatic balance.

3.1. Case 1: Inertia-gravity waves

The nonhydrostatic inertia-gravity wave test involves the evolution of a potential temperature perturbation in a channel with periodic boundary conditions on the left and right boundaries. The initial perturbation radiates to the left and right symmetrically, but because of the mean horizontal flow, does not remain centered about the initial position. The initial conditions we use are identical to those of Skamarock and Klemp [46]; we provide the test case definition below for completeness. The initial state of the atmosphere is taken to have a constant mean flow of $\bar{u} = 20$ m/s in a uniformly stratified atmosphere with a Brunt-Väisälä frequency of $\mathcal{N} = 0.01$/s. Using the definition of Brunt-Väisälä frequency

$$\mathcal{N}^2 = g\frac{\mathrm{d}}{\mathrm{d}z}(\ln\bar{\theta}) \tag{12}$$

yields

$$\bar{\theta} = \theta_0 \mathrm{e}^{\frac{\mathcal{N}^2}{g}z}, \tag{13}$$

where the constant of integration is $\theta_0 = 300$ K. The Exner pressure is obtained from the hydrostatic balance, Eq. (2), as

$$\bar{\pi} = 1 + \frac{g^2}{c_p\theta_0\mathcal{N}^2}\left(\mathrm{e}^{-\frac{\mathcal{N}^2}{g}z} - 1\right). \tag{14}$$

In addition, the potential temperature is perturbed by the amount

$$\theta' = \theta_c \frac{\sin\left(\frac{\pi_c z}{h_c}\right)}{1 + \left(\frac{x - x_c}{a_c}\right)^2},$$

where $\theta_c = 0.01$ °C, $h_c = 10,000$ m, $a_c = 5000$ m, $x_c = 100,000$ m, and $\pi_c = 3.14159265$ is the Archimedes' (trigonometric) constant.

The domain is defined as $(x,z) \in [0,300,000] \times [0,10,000]$ m with $t \in [0,3000]$ s. No-flux boundary conditions are used along the bottom and top boundaries while periodic boundary conditions are used along the lateral boundaries.

3.2. Case 2: Rising thermal bubble

The rising thermal bubble test shows the evolution of a warm bubble in a constant potential temperature environment. Because the bubble is warmer than the ambient air it rises while deforming as a consequence of the shearing motion caused by the velocity field gradients until it forms a mushroom cloud. The initial conditions we use are similar to those of Robert [42] with the air mass initially at rest and in hydrostatic balance. Setting $\bar{\theta} =$ constant in (2) immediately provides for the Exner pressure. To drive the motion of the air, the following potential temperature perturbation is then added:

$$\theta' = \begin{cases} 0 & \text{for } r > r_c, \\ \frac{\theta_c}{2}\left[1 + \cos\left(\frac{\pi_c r}{r_c}\right)\right] & \text{for } r \leqslant r_c, \end{cases}$$

where $\theta_c = 0.5$ °C, π_c is the trigonometric constant, $r = \sqrt{(x - x_c)^2 + (z - z_c)^2}$ with the following constants: $\bar{\theta} = 300$ K, $r_c = 250$ m, and $(x,z) \in [0,1000]^2$ m with $t \in [0,700]$ s and $(x_c, z_c) = (500, 350)$ m. The boundary conditions for all four boundaries are no-flux.

3.3. Case 3: Robert smooth bubble

Test case 3 is similar to the Robert smooth bubble presented in [42]; however, we use a cosine wave instead of the Gaussian used in [42]. The definition of this case is almost identical to the rising thermal bubble except that the domain is now defined as $(x,z) \in [0,1000] \times [0,1500]$ m with $t \in [0,800]$ s and $(x_c, z_c) = (500, 260)$ m. Recall that in case 2, the bubble is initially positioned at $y_c = 350$ whereas now we lower this position and let the bubble evolve for an additional 100 s (from 700 s in case 2 to 800 s in case 3). The longer integration requires us to move the top boundary in order to avoid having the bubble hit this boundary.

3.4. Case 4: Density current

The density current test was proposed in [51] and concerns the evolution of a cold bubble dropped in a neutrally stratified atmosphere. Because the bubble is cold, it sinks eventually hitting the ground. At this point, the bubble begins to shear as it travels along the ground forming Kelvin–Helmholtz rotors. As discussed in [51], viscosity is required in order to obtain a grid-converged solution. The initial conditions for this case are quite similar to those of the rising thermal bubble; however, the differences are in the domain size and the shape of the cold cosine bubble. The potential temperature perturbation is defined as

$$\theta' = \frac{\theta_c}{2}[1 + \cos(\pi_c r)],$$

where $\theta_c = -15$ °C

$$r = \sqrt{\left(\frac{x - x_c}{x_r}\right)^2 + \left(\frac{z - z_c}{z_r}\right)^2}$$

and $r_c = 1$. The domain is defined as $(x,z) \in [0, 25, 600] \times [0, 6400]$ m with $t \in [0, 900]$ s and the center of the bubble is originally positioned at $(x_c, z_c) = (0, 3000)$ m with the size of the bubble defined by the parameters $(x_r, z_r) = (4000, 2000)$ m. The boundary conditions for all four boundaries are no-flux and a dynamic viscosity of $\mu = 75$ m^2/s is used. Note that we have defined only half the domain in the horizontal as proposed in Straka et al. [51].

3.5. Case 5: Schär mountain

The Schär mountain test concerns the steady-state solution of hydrostatic flow over a five-peak mountain chain, with steady inflow and outflow boundary conditions. The initial conditions and mountain profile are given in Schaï et al. [43] which we now provide for completeness. The initial state of the atmosphere is taken to have a constant mean flow of $\bar{u} = 10$ m/s in a uniformly stratified atmosphere with a Brunt-Väisälä frequency of $\mathcal{N} = 0.01$/s. Using the definition of Brunt-Väisälä frequency, Eq. (12) yields the reference potential temperature given in Eq. (13), the Exner pressure given in Eq. (14) where the constant of integration is taken to be $\theta_0 = 280$ K. The domain is defined as $(x,z) \in [-25,000, 25,000] \times [0, 21,000]$ m with $t \in [0, 10]$ h. The mountain is defined as

$$h(x) = h_c \mathrm{e}^{-\left(\frac{x}{a_c}\right)^2} \cos^2\left(\frac{\pi_c x}{\lambda_c}\right)$$

with the parameters $h_c = 250$ m, $\lambda_c = 4000$ m, and $a_c = 5000$ m. This profile is shown in Fig. 1 where the axes have been magnified as follows $(x,z) \in [-10,000, 10,000] \times [0, 1000]$ m in order to show the five peaks of the mountain. No-flux boundary conditions are used along the bottom surface while non-reflecting boundary conditions are used along the top and lateral boundaries. Note that this test is in the hydrostatic range since $\frac{\mathcal{N} a_c}{\bar{u}} > 1$.

3.6. Case 6: Linear hydrostatic mountain

The linear hydrostatic mountain test involves the steady-state solution of linear hydrostatic flow over a single-peaked mountain with constant inflow and outflow boundary conditions. The initial conditions and

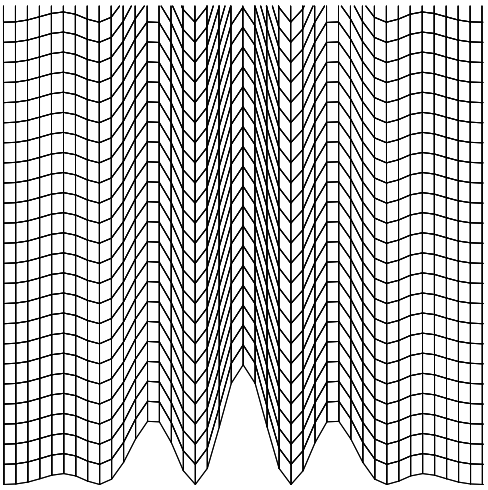


Fig. 1. Case 5: The Schär Mountain profile magnified such that $(x,z) \in [-10,000, 10,000] \times [0, 1000]$ m.

mountain profile are presented in Smith [49] and Durran and Klemp [11]. The initial state of the atmosphere consists of a constant mean flow with $\bar{u} = 20$ m/s in an isothermal atmosphere with constant temperature $\overline{T} = 250$ K. Since in the isothermal case we have $\mathcal{N} = \frac{g}{\sqrt{c_p \overline{T}}}$, the potential temperature and Exner pressure profiles are immediately provided by Eqs. (13) and (14), this latter reducing to

$$\bar{\pi} = \mathrm{e}^{-\frac{g}{c_p \overline{T}} z}.$$

The domain is defined as follows: $(x,z) \in [0, 240, 000] \times [0, 30, 000]$ m with $t \in [0, 10]$ h with the *versiera di Agnesi* mountain profile defined as

$$h(x,z) = \frac{h_c}{1 + \left(\frac{x - x_c}{a_c}\right)^2}, \tag{15}$$

where $h_c = 1$ m, $x_c = 120,000$ m, and $a_c = 10,000$ m. It can be verified that $\frac{\mathcal{N} a_c}{\bar{u}} > 1$, so that the flow is in the hydrostatic range. Non-reflecting boundary conditions are required along the lateral and top boundaries. In addition, no-flux boundary conditions are used along the bottom.

3.7. Case 7: Linear nonhydrostatic mountain

The linear nonhydrostatic mountain test involves the steady-state solution of linear nonhydrostatic flow over a single-peaked mountain with constant inflow and outflow boundary conditions. For this test the initial state of the atmosphere consists of a constant mean flow of $\bar{u} = 10$ m/s in a uniformly stratified atmosphere with a Brunt-Väisälä frequency of $\mathcal{N} = 0.01$/s. Potential temperature and Exner pressure profiles are given by Eqs. (13) and (14), respectively, with $\theta_0 = 280$ K. The mountain profile given by Eq. (15) is used, with the numerical parameters now specified as $h_c = 1$ m, $x_c = 72,000$ m, and $a_c = 1000$ m. The computational domain is $(x,z) \in [0, 144, 000] \times [0, 30, 000]$ m with $t \in [0, 5]$ h. Note that since $\frac{\mathcal{N} a_c}{\bar{u}} = 1$ the flow is then in the nonhydrostatic regime. No-flux boundary conditions are used along the bottom and non-reflecting boundary conditions along the lateral and top boundaries.

4. Spatial discretization of the governing equations

For the sake of brevity, we shall only discuss the discretization of set 3 which we now write in the following compact vector form